

State-of-the-Art Methods for Sequential Data Prediction with Exogenous Inputs

Overview of the Sequential Forecasting Problem

Forecasting a sequential variable y_t (e.g. the Federal Funds Rate) based on its own history and other time-dependent variables $(x_{1,t}, x_{2,t}, \dots, x_{n,t})$ is essentially a **multivariate time-series forecasting** problem. In this setting, the target series is univariate (one main output), but we leverage **exogenous** or **covariate** sequences (other macroeconomic indicators in the Fed Funds example) to improve predictions. Traditional statistical approaches like ARIMAX (ARIMA with exogenous inputs) or Vector Autoregression (VAR) have long been used for such problems. However, these linear models often struggle with complex nonlinear relationships present in economic data ¹ ². This has spurred interest in machine learning and especially **deep learning** methods that can automatically learn both linear and nonlinear dependencies from sequential data.

Classical vs. Machine Learning Approaches

Classical methods: Techniques such as ARIMA (and its variants) and VAR explicitly model temporal correlations but assume linear dynamics. They can incorporate other variables (e.g. ARIMAX includes external regressors), but any nonlinear interactions or regime changes are hard to capture. For example, combinations of ARIMA with neural networks or support vector machines have been tried to address these limitations, often yielding better accuracy than pure ARIMA ². Similarly, hybrid models that decompose a time series (e.g. via wavelet or spectral methods) and then apply linear models and neural nets have shown improved performance in complex economic and energy price forecasting ³.

Machine learning methods: On the other hand, modern ML models can automatically fit complex patterns. Tree-based models (Random Forests, Gradient Boosting) and classical neural networks (multilayer perceptrons) can take lagged values of y_t and other variables as features. These often outperform naive statistical baselines on large datasets, but they don't inherently account for sequential structure. The real leap has come from **deep learning sequence models** – particularly **Recurrent Neural Networks (RNNs)** and more recently **Transformer** architectures – which are now state-of-the-art for many time-series forecasting tasks ⁴ ⁵. These models treat the data as sequences, inherently capturing temporal dependencies and patterns (trends, seasonality, regime shifts) that simpler models might miss.

RNN-Based Models (LSTM/GRU) for Time Series

Recurrent Neural Networks process sequential data by maintaining a hidden state that evolves over time, making them well-suited to time-series. Two popular RNN variants are **Long Short-Term Memory (LSTM)** and **Gated Recurrent Units (GRU)**, which were designed to combat the vanishing gradient problem and capture long-term dependencies. These models can naturally ingest multivariate input sequences (feeding in $[y_t, x_{1,t}, \dots, x_{n,t}]$ at each time step) and learn how those variables jointly influence the future

target. Over the past decade, LSTMs and GRUs have been successfully applied to economic and financial data forecasting, often outperforming both traditional models and other machine learning methods ⁴ ⁶. For instance, research has shown that an LSTM can **outperform ARIMA and exponential smoothing** in predicting macroeconomic indicators like GDP or inflation ⁷ ⁸, and even **beat univariate statistical models in interest rate forecasting** ⁴. RNN-based models are effective because they can model nonlinear dynamics and retain memory of past states; indeed, studies found they “*better capture the implicit decision rules*” of complex processes like central bank policy compared to linear rules-based models ⁹.

Enhancements to RNNs: In practice, RNN models are often improved by adding mechanisms for greater flexibility. One such enhancement is the **attention mechanism**, which lets the network learn to weigh the importance of different past time steps or different variables when making a prediction. Incorporating attention into RNNs has improved accuracy by focusing the model on the most relevant parts of the history (for example, paying special attention to recent changes in inflation or employment when forecasting interest rates). In fact, recent “hybrid” RNN architectures combine LSTM/GRU with attention layers to boost performance ⁵. An example is the **Temporal Fusion Transformer (TFT)** introduced by Lim et al. (2021), which uses LSTM layers for local sequential processing together with an attention-based “fusion” module to select important features and time steps ¹⁰. TFT was shown to achieve high accuracy on multi-horizon forecasts while also providing interpretability (e.g. revealing which economic covariates were most influential at different forecast horizons) ¹⁰. This represents a bridge between pure RNN approaches and the Transformer models discussed next.

Transformer-Based Models for Sequential Data

Transformers have revolutionized sequence modeling in recent years by relying entirely on **self-attention** mechanisms instead of recurrence ¹¹. Originally developed for NLP, the Transformer architecture (Vaswani et al. 2017) can capture long-range dependencies in a sequence more effectively and in parallel, which is very appealing for time-series tasks that may have long-term seasonal patterns or delayed effects. When adapted to time-series forecasting, Transformers have demonstrated state-of-the-art performance, especially for long sequence prediction and when incorporating multiple correlated time series ¹¹ ¹². A key advantage is that **self-attention** can flexibly attend to any past time step (or any explanatory variable) that is relevant for predicting the future, rather than fixed-size “windows” or purely recent lags as in standard RNNs.

Notable Transformer-based architectures for time series (2019–2024) include:

- **Informer (Zhou et al., 2021):** Introduces a **ProbSparse self-attention** mechanism that selects the most informative query-key pairs, reducing the quadratic complexity of vanilla attention. This makes Informer much more efficient for **long-sequence forecasting**, enabling it to handle longer input histories without blowing up computation ¹³. Informer also adds **distillation** on the attention outputs to further shorten sequences, all while maintaining prediction accuracy for long horizons ¹⁴.
- **Autoformer (Wu et al., 2021):** Builds on Transformers with a novel **series decomposition** idea. It uses an **Auto-Correlation mechanism** instead of standard attention to more directly capture seasonal patterns and long-term trends in the data ¹⁵. By progressively decomposing the time series and modeling the remainder, Autoformer can handle complex temporal patterns more

effectively while being computationally efficient ¹⁵. This approach improved long-horizon forecast accuracy and won accolades in long-term forecasting benchmarks.

- **ETSformer (Woo et al., 2022):** A hybrid approach that **integrates exponential smoothing (ETS) concepts** into the Transformer architecture ¹⁶. ETSformer's layers explicitly decompose the input into level, trend, and seasonality components (mimicking classical Holt-Winters exponential smoothing) and process each with attention. By blending statistical time-series components with deep learning, ETSformer achieves strong accuracy and provides more interpretable components of the forecast ¹⁶. It addresses one weakness of black-box models by preserving interpretability akin to classical models, while still leveraging Transformer's power.
- **Temporal Fusion Transformer (TFT, 2020):** Mentioned earlier, TFT is an attention-based **hybrid** that includes recurrent blocks. It's notable for handling not only past observed values but also known future inputs (like planned policy changes or calendar effects) through specialized attention heads, and for producing **interpretable attention weights** highlighting each feature's importance ¹⁰. TFT has been applied successfully to a variety of economic forecasting tasks and is considered a **state-of-the-art multivariate forecasting model** in its class.
- *(Many other Transformer variants exist, such as Reformer (efficient attention via locality-sensitive hashing), NSTransformer (non-stationary transformer tackling time series with changing statistics), FEDformer, PatchTST, etc. The rapid proliferation of these models in the last few years underscores how active this research area is ¹⁷ ¹³.)*

Overall, Transformer-based methods have become **cutting-edge for sequential prediction problems**, especially when there are long-range dependencies or multiple related series. They often **outperform earlier RNNs** on benchmarks, given enough data ⁵. Importantly, they also excel at incorporating *exogenous covariates*: the attention mechanism can naturally weight the influence of different variables at each time step. For example, a Transformer model can learn to pay more attention to the inflation rate series or employment figures whenever those become critical for predicting an interest rate move. Empirical studies show that including such explanatory variables in Transformer models yields significant accuracy gains – one study reported up to a **12.4% reduction in forecast error (NRMSE)** when relevant exogenous features were added to a Transformer-based predictor ¹⁸ ¹⁹. This ability to seamlessly fuse information from many sources is a major reason Transformers are now a go-to solution for complex time-series forecasting.

Hybrid and Ensemble Models

Rather than relying on a single architecture, **hybrid models** aim to combine the strengths of multiple approaches. The rationale is that time series often have both linear/stationary components and nonlinear patterns; no single model may capture all aspects perfectly. By hybridizing, we can tackle different facets of the data with specialized tools. This has a rich history in forecasting:

- **Classical-ML Hybrids:** As noted, researchers have combined ARIMA (to handle linear trends/cycles) with machine learning models (to capture residual nonlinearities). For example, a classic hybrid by Zhang (2003) used ARIMA to model linear structure and a neural network to model the remaining pattern, yielding better accuracy than either alone ². Others have paired ARIMA with SVMs ² or

with ANN/RNNs ¹, and found similarly improved results in stock price and energy demand forecasting.

- **Deep Learning Hybrids:** Within deep learning, we see combinations like **CNN + LSTM** (a CNN to extract short-term local trends or seasonality, feeding into an LSTM for longer-term dependencies) ²⁰, or **CNN-BiLSTM with attention (ECA)** for financial forecasting ²¹. Each component addresses a different aspect: CNNs are good for picking up local patterns or sudden changes, while LSTMs excel at sequential dependencies; attention modules can then highlight the important signals. Empirical evidence strongly suggests that such *composite models outperform their single-component counterparts*, especially on noisy financial time series ²². In one example, a CNN-BiLSTM-ECA model outperformed standalone CNN or LSTM models on multiple Chinese stock indices ²³. Likewise, an ensemble of ARIMA with CNN-LSTM was shown to beat each individual model on Bitcoin and other datasets ²⁴.
- **LSTM + Transformer Hybrids:** A very recent innovation (2024–2025) is to combine recurrent networks with transformers into one unified model. The idea is to use RNN modules to efficiently capture local sequential structure, and a Transformer module to capture long-range dependencies and interactions among variables. *This effectively merges the two leading paradigms in sequence modeling.* A 2025 study by Kabir *et al.* proposed an **LSTM-Transformer-MLP ensemble model** (dubbed LSTM-mTrans-MLP) for financial time series forecasting ²⁵. In this model, an LSTM encoder first learns short- and medium-term patterns, then a “modified Transformer” layer (with a custom attention mechanism) further refines the representations, and finally a feed-forward MLP layer produces the forecast ²⁵ ²⁶. This hybrid achieved **state-of-the-art accuracy** on multiple market datasets (stock indices, cryptocurrencies), outperforming baseline models like pure LSTM, pure CNN, and even other hybrids like CNN-LSTM ²⁷ ²⁸. The authors note that combining LSTM and Transformer in an encoder–decoder structure yielded a robust model that generalized well across different types of financial series, whereas previous works often had to retune or redesign models for each dataset ²⁹. In summary, the trend is towards *modular deep learning architectures* that can capture different aspects of the data’s behavior – these often end up being more **accurate and resilient** than single-network models ²⁹ ³⁰.
- **Ensembles:** Another form of hybridization is ensembling multiple models. Instead of one architecture, one can train several different models (say, an LSTM, a Transformer, and a regression) and then combine their predictions (e.g. by averaging or through a meta-learner). This can reduce variance and capture complementary strengths. For example, researchers have used boosting methods (like AdaBoost) to ensemble networks (each “weak learner” being an LSTM or GRU) for forecasting exchange rates ³¹. Such ensemble approaches can yield more stable and accurate forecasts, though at the cost of increased complexity.

Latest Developments and Outlook (2024–2025)

The field of sequential forecasting is evolving rapidly, and **state-of-the-art methods** continue to improve. A notable recent direction is the rise of **large pretrained models for time series**, analogous to how large language models (LLMs) transformed NLP. These are sometimes called **Time-Series Foundation Models (TSFMs)** or **Time-Series Language Models (TSLMs)** ³². The idea is to train Transformer-based architectures on massive collections of time-series data (potentially across many domains) to learn general patterns, and then fine-tune or prompt them for specific forecasting tasks. For instance, models like

TimeGPT, LagLlama, Time-LLM, and others were introduced in 2023–2024 ³². Early studies indicate that such models can **achieve accuracy gains** in forecasting by transferring knowledge across domains ³². In the macroeconomic realm, researchers have started to explore using large language models (like GPT-3/4) to forecast economic indicators. One experiment found that a big pretrained model (Google’s PaLM, used in a zero-shot manner) could generate **inflation forecasts** that were competitive with – and often even better than – those of professional human forecasters over longer-term horizons ³³. Another 2025 study applied a GPT-based approach to predict the Federal Funds Rate and noted that the LLM slightly outperformed specialized LSTM/GRU models in recent volatile periods ³⁴ ³⁵. These results are preliminary but suggest a promising avenue where *foundation models* capture extremely complex relationships (including possibly interpreting news or textual FOMC statements alongside numeric data) to inform forecasts ³⁶ ³⁷.

In parallel, there is ongoing research into improving **forecasting under uncertainty** and **probabilistic predictions**. Modern techniques often output not just point forecasts but full predictive distributions. Transformer-based probabilistic models (e.g. forecasting a range of possible outcomes with confidence intervals) have been studied, and incorporating exogenous variables in these models improves not only point accuracy but also the reliability of uncertainty estimates ³⁸ ¹⁹. This is particularly relevant for policy rates like the Fed Funds Rate, where assessing the risk of different scenarios (high vs. low inflation outcomes) is as important as predicting the baseline case.

In summary, the consensus in 2025 is that **deep learning approaches, especially Transformer-based architectures, represent the state of the art** for sequential data prediction with multiple input series. These models have shown superior accuracy in capturing the complex, nonlinear, and long-term dependencies in macroeconomic data compared to traditional models ⁴. The best practice is often to include all relevant explanatory time series (e.g. inflation, GDP growth, unemployment, etc. for interest rate forecasting) and let the model’s attention or learned weights figure out how each one contributes to the future trajectory of the target ¹⁸ ¹⁹. Furthermore, **hybrid strategies** (combining neural architectures, or blending neural nets with domain-specific components) are proving to be highly effective, as they can exploit different patterns in the data simultaneously ²² ³⁰. As new research emerges, we are seeing continual improvements – from tailored Transformer variants to large multi-domain models – all pushing the frontier of sequential prediction. For a problem like forecasting the Fed Funds Rate with other economic indicators, a reasonable cutting-edge solution today might be a Transformer-based model (for its ability to handle sequences and multiple covariates) possibly enhanced with a hybrid framework (to incorporate any known structures or to ensemble with an RNN). This approach would leverage the latest advances and is supported by numerous studies demonstrating its superior performance in similar forecasting tasks ²⁸ ³³.

Sources:

- Lim, B. *et al.* (2021). *Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting*. Int. J. Forecast. – Introduces the TFT model combining LSTM and attention for multivariate forecasts ¹⁰.
- Zhou, H. *et al.* (2021). *Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting*. – Proposes the Informer architecture with sparse self-attention for long-range time series forecasting ¹³.
- Wu, H. *et al.* (2021). *Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting*. – Develops the Autoformer model using series decomposition and auto-correlation mechanisms ¹⁵.

- Woo, S. et al. (2022). *ETSformer: Exponential Smoothing Transformers for Time-series Forecasting*. – Integrates exponential smoothing into Transformer, improving long-term forecasts by modeling level/trend/seasonality components ¹⁶ .
- Kabir, M.R. et al. (2025). *LSTM-Transformer-Based Robust Hybrid Deep Learning Model for Financial Time Series Forecasting*. *Sci*, 7(1), 7 – Demonstrates a hybrid of LSTM, modified Transformer, and MLP achieving SOTA results on various financial time series ²⁸ ³⁰ .
- Caetano, R. et al. (2025). *Transformer-Based Models for Probabilistic Time Series Forecasting with Explanatory Variables*. *Mathematics*, 13(5):814 – Evaluates multiple Transformer variants on retail data, finding that incorporating exogenous variables yields ~12% error reduction ¹⁸ ¹⁹ .
- Chan-Lau, J. et al. (2025). *Forecasting Federal Funds Rates with AI: LSTM, GRU, and Large Language Model Approaches*. AMRO Working Paper 25-10 – Finds that deep LSTM/GRU models outperform traditional models for Fed rate prediction, and a large GPT-based model can even slightly outperform those RNNs in recent period ⁴ ³³ .
- (Additional background) Various authors (2022–2024) on hybrid and ensemble models in financial forecasting – e.g. CNN-LSTM and ARIMA hybrids ¹ ²⁴ , attention-enhanced RNNs ⁵ , and time-series foundation models ³² , reflecting the ongoing advancements in the field.

¹ ² ³ ²⁰ ²¹ ²² ²³ ²⁴ ²⁵ ²⁶ ²⁷ ²⁸ ²⁹ ³⁰ LSTM-Transformer-Based Robust Hybrid Deep Learning Model for Financial Time Series Forecasting

<https://www.mdpi.com/2413-4155/7/1/7>

⁴ ⁵ ⁶ ⁷ ⁸ ⁹ ³¹ ³³ ³⁴ ³⁵ ³⁶ ³⁷ amro-asia.org

https://amro-asia.org/wp-content/uploads/2025/09/AMRO_WP_25_10_FedAI_model.pdf

¹⁰ ¹¹ ¹² ¹³ ¹⁴ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ ³⁸ Transformer-Based Models for Probabilistic Time Series Forecasting with Explanatory Variables

<https://www.mdpi.com/2227-7390/13/5/814>

³² Macroeconomic Forecasting with Large Language Models The views and opinions expressed here are those of the authors and do not necessarily reflect the views or positions of any entities they are affiliated with.

<https://arxiv.org/html/2407.00890v1>